GIS Practicum

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Part 1. Small Data, Big Data, and AI in Geography(AAG)

Core Tension: Data Quantity vs. Data Quality

In geography and GeoAI, **more data does not necessarily mean better data**.

- **Big Data**
  - Emphasizes: volume, coverage, and heterogeneity
  - Sources: remote sensing, street-view imagery, sensors, social media
  - Challenges:
    - High noise levels
    - Label inconsistency
    - Semantic ambiguity

If data quality is insufficient, the value of big data is significantly reduced.

- **Small Data**
  - Emphasizes: **high quality, high relevance, and high signal-to-noise ratio**
  - Typically includes:
    - Expert-labeled datasets
    - Task-specific curated data
    - Domain-informed samples

Key insight:

“Less but better” data can be more effective than “more but noisier” data.

Paradigm Evolution: From Small Data to Big Data to Agent-based AI

This progression can be conceptualized as a **three-stage evolution of AI paradigms in geography**:

Stage 1: Small Data Paradigm

Keywords: Fine-tuning, task-specific learning

- Relies on limited but high-quality datasets
- Adapts pre-trained models via **fine-tuning**
- Focus:
  - Task adaptation
  - Domain knowledge injection

Example (my work):

- Disaster damage assessment
- Bi-temporal street-view analysis
- CLIP-based fine-tuning

Essence:

Using small, high-quality data to make models specialized and task-aware.

Stage 2: Big Data Paradigm

Keywords: Foundation models, world models

- Trains models on large-scale datasets
- Learns **general representations** across tasks
- Examples include large multimodal models and world-scale geospatial models (e.g., AlphaEarth)

Characteristics:

- Strong generalization ability
- Cross-task transferability
- Global spatial pattern learning

Key challenges:

- Data quality control
- Bias and representativeness
- High computational cost

Essence:

Using large-scale data to build models with general intelligence and global understanding.

Stage 3: Post-Foundation Paradigm (Zero-shot + Agent-based AI)

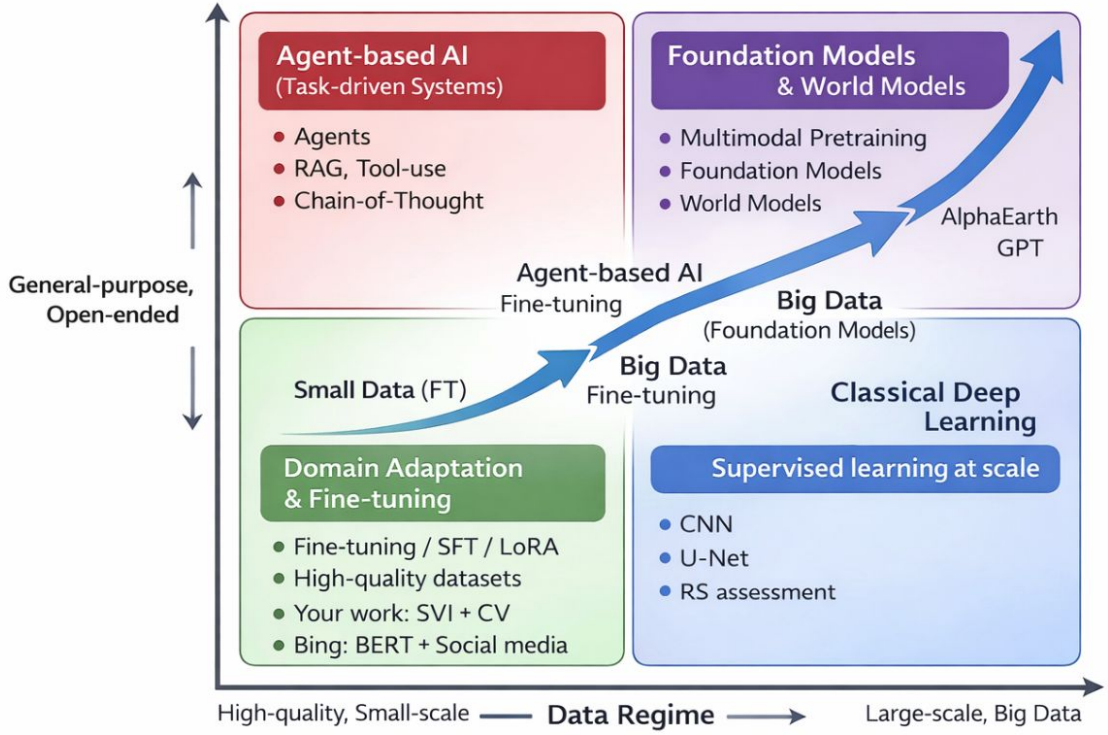
Keywords: Zero-shot learning, multi-agent systems, autonomous AI

- Reduces reliance on labeled data
- Often eliminates the need for task-specific fine-tuning
- Leverages:
  - Zero-shot reasoning
  - Multi-agent collaboration
  - Task decomposition and orchestration

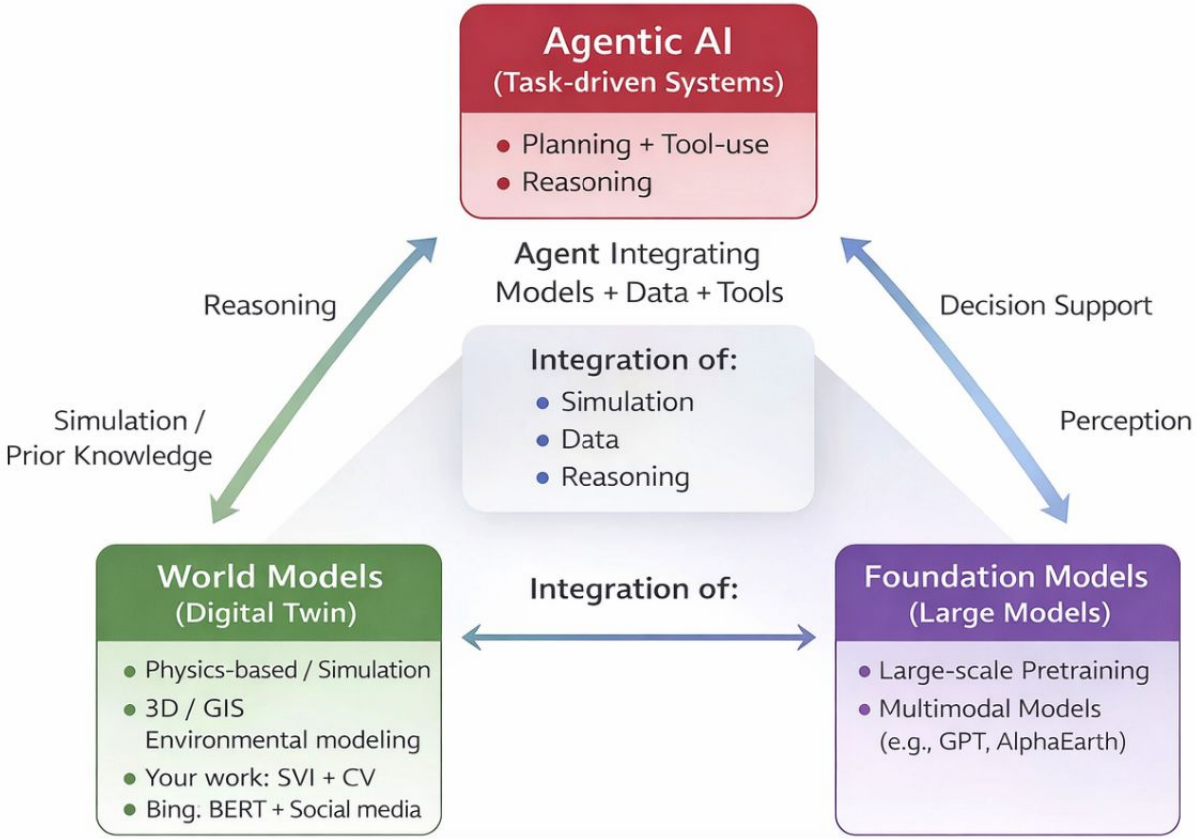
Example:

- GeoAgent4Disaster
- RAPIDMap
- Autonomous GeoAI frameworks

Small Data shapes **what to solve and how to evaluate**, while Big Data determines **what can be perceived and generalized**—together enabling agent-driven GeoAI systems.



A Unified GeoAI Paradigm:
From Models to Agents



General Advice

1. Research questions are the most compelling part of geography.

Human activities and environmental processes are constantly evolving, which means there will always be new questions to explore. What truly matters is not just methods, but the ability to identify meaningful geographic problems.

2. Don't underestimate your programming ability, but don't overestimate your engineering ability.

In the age of AI, implementation barriers are lower, but engineering experience remains invaluable. Software engineers will not be replaced, because the accumulated knowledge from building, debugging, and iterating systems is irreplaceable.

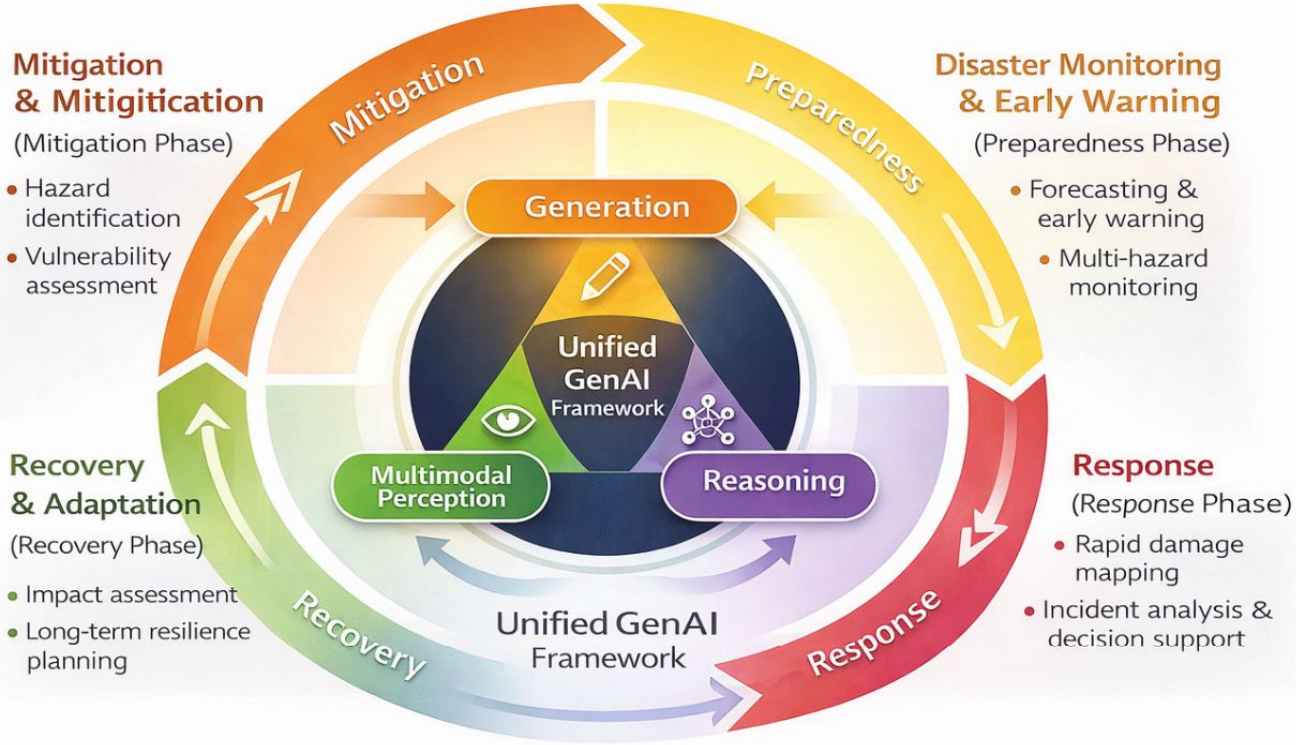
3. Embrace technological upgrades early.

New methods and tools not only improve performance but also reshape workflows and significantly increase research efficiency. The earlier you adopt them, the faster you develop strong technical intuition.

4. Doing brings you closer to completion, but thinking is your foundation as a PhD.

Execution moves you forward, but thinking determines your direction. As a PhD student, never forget that your ability to think critically is the core of your long-term value.

Generative AI for Disaster Resilience Tasks



Mitigation & Mitigitication
(Mitigation Phase)
- Hazard identification
- Vulnerability assessment

Disaster Monitoring & Early Warning
(Preparedness Phase)
- Forecasting & early warning
- Multi-hazard monitoring

Recovery & Adaptation
(Recovery Phase)
- Impact assessment
- Long-term resilience planning

Response
(Response Phase)
- Rapid damage mapping
- Incident analysis & decision support

Generation
Data enhancement & scenario simulation

Multimodal Perception
Integration of multimodal geospatial data

Reasoning
Cross-scale spatial-temporal reasoning

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TEXAS A&M UNIVERSITY

Spatial Data Science

Understanding Spatial Patterns, Processes, and Scales

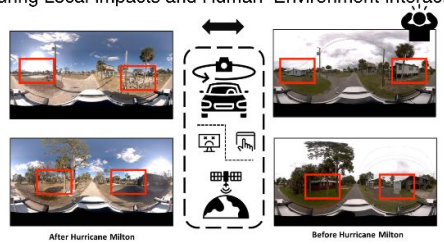
Representation

Recognition

Geospatial AI

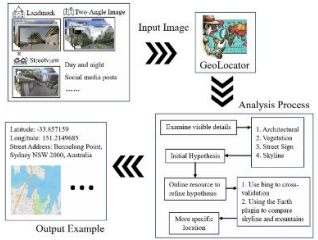
Multimodal Ground-Level Analysis-

Capturing Local Impacts and Human–Environment Interactions



Geospatial Reasoning-

Integrating Spatial Knowledge for Geographic Understanding



Reasoning

Prediction

Autonomous GeoAI-

Toward Proactive, Responsible, and Trustworthy GeoAI

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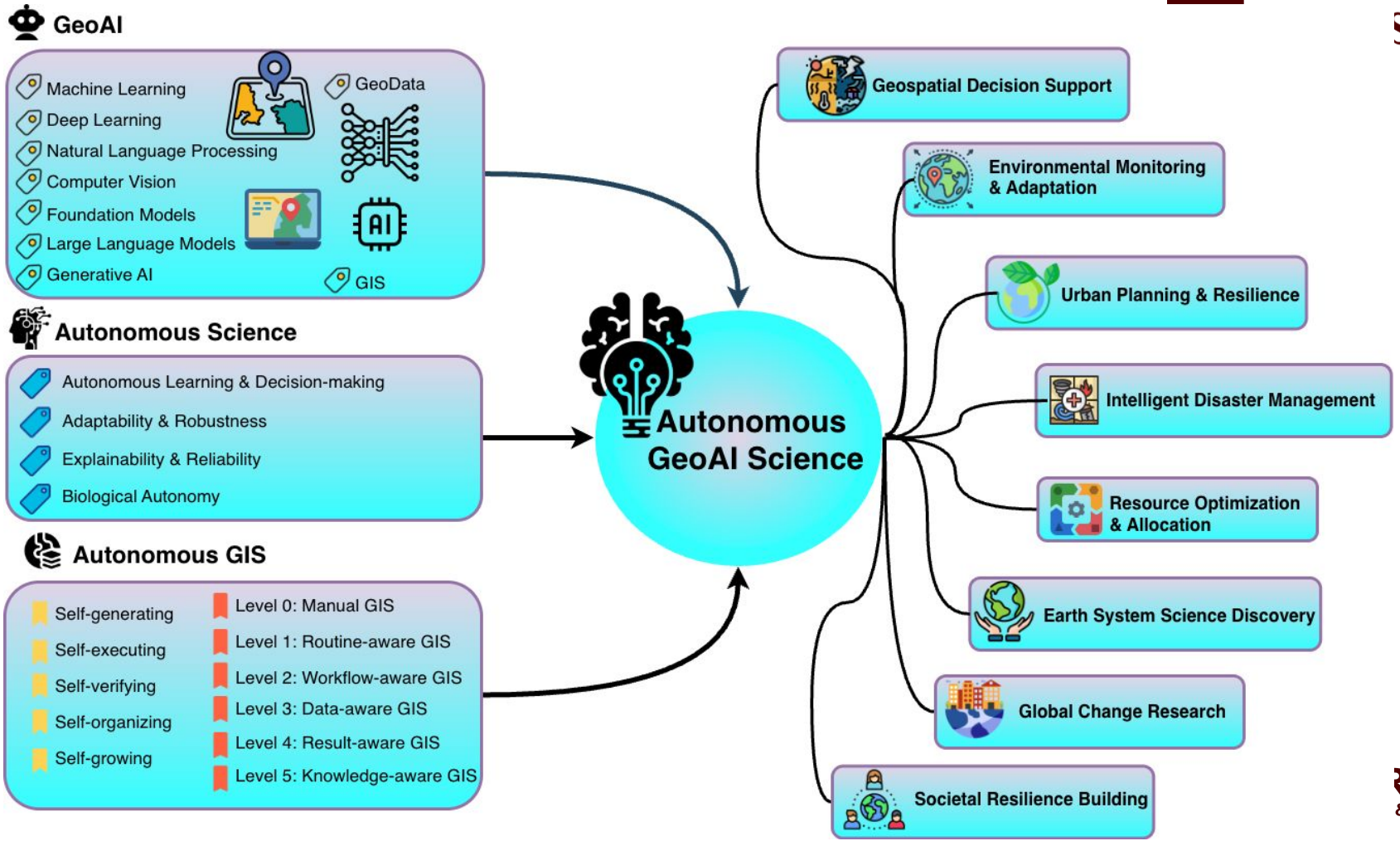

GeoAI
Machine Learning
Deep Learning
Natural Language Processing
Computer Vision
Foundation Models
Large Language Models
Generative AI
GeoData
GIS
Autonomous Science
Autonomous Learning & Decision-making
Adaptability & Robustness
Explainability & Reliability
Biological Autonomy
Autonomous GIS
Self-generating
Self-executing
Self-verifying
Self-organizing
Self-growing
Level 0: Manual GIS
Level 1: Routine-aware GIS
Level 2: Workflow-aware GIS
Level 3: Data-aware GIS
Level 4: Result-aware GIS
Level 5: Knowledge-aware GIS
Autonomous GeoAI Science
Geospatial Decision Support
Environmental Monitoring & Adaptation
Urban Planning & Resilience
Intelligent Disaster Management
Resource Optimization & Allocation
Earth System Science Discovery
Global Change Research
Societal Resilience Building

Evolution of Focus: From Capability to Institutional Trust & Responsibility

Stage 1. Tool-Centric GeoAI (Technology-Driven)

Focus: Utilizing specific technologies (ML, DL, GIS) as tools to assist human analysis.
Keywords: Efficiency, Automation of tasks, Human-in-the-loop.

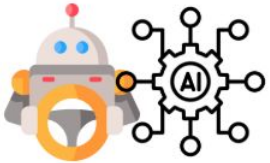
Stage 2. Data-Centric GeoAI (Big Data-Driven)

Focus: Leveraging large-scale geospatial data acquisition, storage, and processing capabilities.
Keywords: Big Data, Cloud Computing, Distributed GIS, Sensor Networks.

Stage 3. Automated GeoAI (Perception-Driven)

Focus: Rapid perception and automated monitoring using AI, but with limited reasoning abilities.
Keywords: Real-time monitoring, Early warning, Pattern recognition, Perception.

Stage 4. Autonomous GeoAI (Capability-Driven)



Focus: High-level autonomous capabilities in perception, reasoning, and decision-making.
Keywords: Self-learning, Autonomous GIS, Generative AI.

Stage 5. Fiduciary GeoAI (Institutional Trust Paradigm)

Focus: Embedding autonomous capabilities within definable institutional frameworks and value constraints.
Keywords: Delegated Autonomy, Auditability, Accountability Alignment.

Note: This figure illustrates the comprehensive evolution of GeoAI, highlighting the critical paradigm shift from autonomous capabilities to institutionally embedded fiduciary responsibility.

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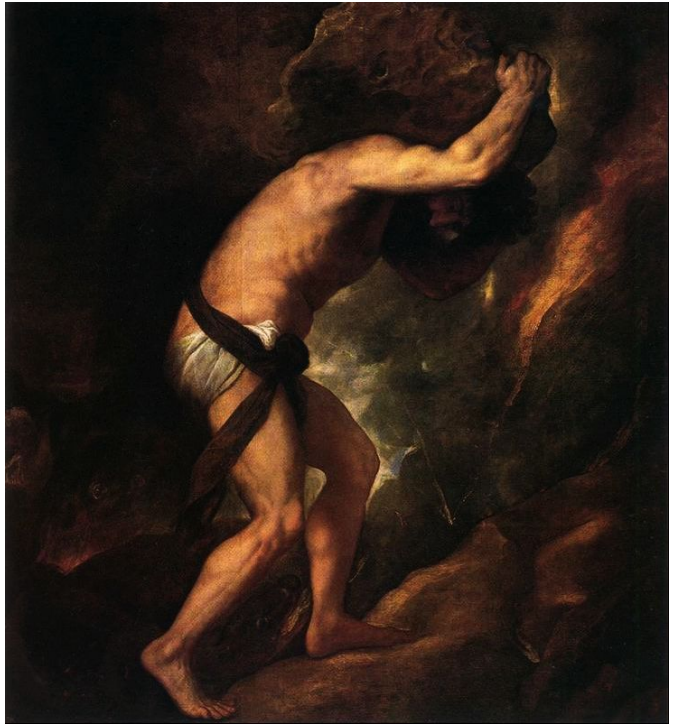
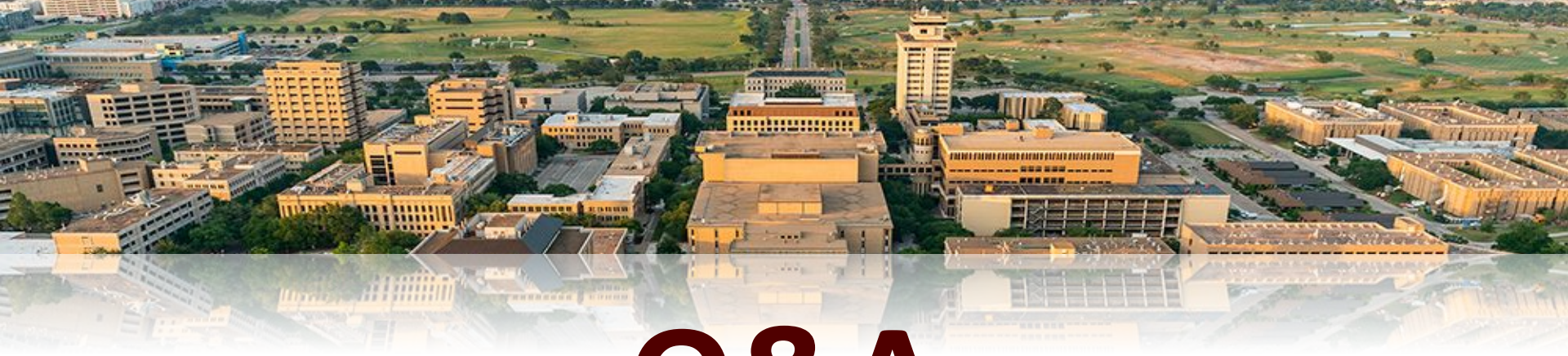
Final Part. Sisyphus and the Philosophy of the Absurd



In Camus' philosophy of the absurd:

Sisyphus is condemned
to push a rock up a mountain forever.
Each time it reaches the top,
it rolls back down.
This cycle repeats endlessly, without inherent meaning.



Q&A

Welcome any comments/questions

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